

Escapement System

Starting with a prototype, I modeled the escapement mechanism editing the stiffness of the hairspring and weight of the balance wheel. In the real design, the escapement is powered using a steel torsion spring and scaled down to fit inside the housing.



As seen above, the connections within the clock were designed to be modular to make iterating and failing easier. There were multiple escape wheels and hairsprings that needed to be redesigned to reach the final.

Inside and Outside



Above is the gearing for the clock with every gear 3D printed and spinning on bearings for the least friction possible.

To the right is the back and front of the clock, which shows features like:

- The blue cover plates
- The ratchet torsion spring winder
- The hour to minute gear ratio
- The modular assembly style
- The balance wheel weights

